

Multi-Modal Vision Based Tactile Sensor Dataset Creation

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Abstract—This research focuses on collecting data with the Daimon-Tac WS Vision-Based Tactile Sensors (VBTS) for the purpose of material classification. The investigation of VBTS, particularly the Daimon-Tac WS, aims to clarify the potential of simple-marker-based transduction principles for material classification.

The research in this report focuses on testing the Daimon sensor by sliding it across different orientations of various materials. Following the completion of data collection, a comparative analysis is conducted within each material to evaluate the difference between the direction of the sensor's movements and to determine whether the sensor successfully generated a "fingerprint" for each material. A further comparison is conducted among the various materials to evaluate their key distinctions.

The collected data will expand the currently limited dataset available for the Daimon sensor. This data collection will enhance the training of the neural network to classify materials with more accuracy. The application of data collection for material classification aids in object manipulation, enhances assistive technology, and improves remote teleoperation and training.

Index Terms—Daimon Sensor, VBTS, material classification, data collection.

I. INTRODUCTION

The Daimon W2 integrates the benefits of multi-feature perception and high resolution. The sensor's properties allowed experimenters to determine the contact state and slip detection and evaluate the material's surface roughness and hardness.

The combination of different contact loads, sliding velocities, and material orientations facilitates the collection of extensive datasets for material classification. Material classification is applicable in architectures where robotic systems attain precise and secure object manipulation, in VR/AR teleoperation systems, and in assistive prosthetic devices. [3]

This research study establishes the collection of shear, depth, shear velocity, and depth velocity from the Daimon sensor as the foundation for differentiating materials using slip detection evaluations. The study examines the distinctions among three materials: plastic, metal, and rubber. Differences will be assessed, and the datasets will be fed into a neural network. The algorithm's accuracy, derived from a minimal dataset and materials, must be evaluated when assessing the potential for real-time classification using the trained model.

II. RESEARCH PROBLEM:

Tactile perception denotes the subjective experience produced by mechanoreceptors in the skin upon contact with a surface. To establish a differentiation between materials, the primary distinctions must be analysed. These distinctions include surface roughness and material hardness. By integrating these essential identifiers with sliding velocity and applied pressure, a comprehensive array of information may be extracted for material classification.

The current research lacks the experimental data gathered with the Daimon sensor. The way the Daimon sensor collects data is based on the marker-based transduction principle. This is key in understanding the limitations and how it alters the collection of the materials' "fingerprint" and requires a deeper understanding of how to extract surface roughness and material hardness of the tested materials.

Therefore, this study focuses on utilising the two modalities offered by the Daimon sensor to conduct experiments on three distinct materials to find the distinguishing features. These features are the difference between the surface roughness and hardness of the test materials. The collected data will then be fed into a neural network. This research aims to determine if sufficient information was collected to accurately classify materials during a live classification test.

III. STATE OF THE ART

The available research papers explore a diverse range of vision-based tactile sensors (VBTS) and their production. The topic of tactile perception is also analysed through the use of bio-inspired sensors that connect human perception to robotics. All these academic papers contribute to a better understanding of how to conduct experiments using VBTS and what information should be considered and better understood.

The research papers are available using VBTS, such as those available using the GelSight sensor. Researchers have already collected a substantial number of datasets using the GelSight sensor. These datasets encompass a diverse array of materials, and the current research papers, such as that explored by Shuvo et al. (2025) [12], add a further thirteen materials with varying degrees of softness, hardness, and texture. The GelSight technology captures fine details of the

materials with a single touch, and due to its technology, it relies on image processing and different neural network (NN) approaches for processing and training datasets. The continued research focused on incorporating new materials by applying traditional image processing and machine learning methods to identify key material features. For deep learning methods, they used pre-trained models and adjusted them to the various types of data through transfer learning. They also trained the data by altering the images given through scaling, cropping, and rotating the image. [12]

Research papers, including those by Ding et al. (2017) [13], draw inspiration from the mechanoreceptors in human skin to examine tactile perception. Utilising an artificial finger, they established the correlation between friction-induced vibration signals and the mechanical properties of diverse materials. A classification of materials was established based on data obtained from their physical properties, including surface roughness and material hardness.

The frictional and vibrational effects during the sliding motion of the artificial finger (bio-tactile sensor) on eight different materials with varying hardness and manufactured roughness were evaluated, providing insights into the material's surface roughness and hardness. The overall results showed that a simple touch was insufficient for discerning distinguishing characteristics among various materials, but a sliding motion with differing velocities and applied pressures produced enhanced accuracy in classifying the materials. [13]

A distinct collection of research papers, including those discussed by Zhang et al. (2022) [14], adopts an alternative perspective by emphasising the advancement of tactile sensors, encompassing their fabrication methods, utilised materials, and a comparative analysis of various manufacturing techniques for tactile sensor development. This encompasses various hardware types, preparation procedures, materials, and sensor cameras.

The comprehensive assessment concentrated on the subsequent aspects: Material resilience under prolonged stress exposure. Assemblability that guarantees stability and minimises recognition discrepancies. The elastic deformation of the sensor's material significantly influences sensitivity, thus affecting the efficacy of information acquisition. Multimodality emphasises the collection of diverse information that modalities offer, including tangential and normal deformation, and that is dependent on the manufacturing method, camera, and illumination. [14]

Additional significant research articles, including those conveyed by Li et al. (2025) [3], highlight the operational mechanics of the Daimon sensor, which employs LEDs, a camera, markers, and a rigid yet flexible elastomer pad to facilitate its multimodality. This study compares the technologies of Daimon sensors, which apply the marker-based transduction concept, with those utilising the intensity-based transduction principle, emphasising the distinct advantages and limitations that must be taken into account when conducting and assessing experimental results. Advantages included compatibility with

neural networks, ease of manufacture, and simple structure. Yet, a disadvantage is the lack of sensitivity to spatial texture.

The research, explored by Liao et al. (2025) [4], utilised the Daimon sensor to assess fruit ripeness for automated agricultural systems. The objective of the paper was to assess the reliability of the data supplied by the sensor and to migrate conventional robotic picking processes to include VBTS, facilitating less invasive techniques and minimising contact when assessing fruit maturity. They utilised a tactile continual learning network founded on generative replay for the identification of object hardness. Their attention was directed towards the normal and shear force maps, as well as the optical flow maps generated by the Daimon sensor. To evaluate hardness, they implemented a constant-rate condition whereby a deviation in the derivative of the normal force indicates intrinsic material qualities (ripeness of the fruit). The examination of the fruits ranged from sensitive to firm, facilitating a clearer visual distinction in normal force with time, with firmer fruits exhibiting a larger graph compared to softer varieties of fruit. They maintained a uniform grabbing distance and progressive intervals. They performed numerous experiments to gather data to assess the correlation between the sensor information and the actual ripeness of the fruit. Their findings are encouraging and provide significant improvements in safety and adaptability for robotic applications in automated agriculture systems. [4]

The previously stated application of Daimon sensors in autonomous agriculture mostly concentrated on the normal force and its derivative.

This research article will integrate the aforementioned papers to validate the selected experimental procedures. The study involves applying a specific force and velocity by the sensor while it slides over four distinct orientations of different materials. The results acquired from the Daimon sensor will be input into a neural network. This will address the lack of datasets accessible for material classification using the Daimon sensor.

IV. RESEARCH METHODOLOGY

A. Research Question and Hypotheses:

To conduct experiments, it is imperative to understand the desired data gathered. This process stems from distinguishing materials via their surface roughness and hardness. To accomplish accurate material classification, several experiments need to be conducted.

These experiments need to consider the aforementioned research papers and how to incorporate tactile perception with VBTS technology. For this set of experiments, it lead to defining constant variables such as the sensor sliding velocity and applied pressure during the motion across the material for material classification.

The data collected via the sliding motion will be sufficient for material classification when real-time classification is operated under the same conditions as when data were collected.

B. Classification Architecture:

Following the completion of multiple tests on the materials illustrated in Figure 1, the collected data must be input into a neural network. The selected neural network will be the time-series transformer. The time-series transformer is advantageous for distinguishing critical differences among shear, depth, tangential velocity, and shear velocity for each material.

The neural network was provided with the complete dataset and categorised its results into three material classifications: plastic, metal, and rubber. A diverse dataset will lead to a more precise classifier during the evaluation of the test data.

V. EXPERIMENTAL PROCEDURES

The experiments conducted will incorporate the information on tangential and normal deformation (from the Daimon sensor), and the calculated rate of tangential and depth deformation. Three different materials were tested, and after sixty repetitive swiping actions are completed, the object's orientation will be changed with respect to the sensor.

To ensure repeatability of every swipe, Baxter, manufactured by Rethink Robotics, has been used to conduct the experiments. The experiment was conducted by attaching the sensor to Baxter's gripper, and through the command of Cartesian coordinates, Baxter would move in a swiping motion. To control the speed, the user could enter how long the swiping motion should last. The choice of duration slows down Baxter's movement through joint-space waypoints. It is important to note that the applied pressure was inconsistent during the swiping motion.

The results collected over the sixty repetitive actions by the Daimon sensors were fed into a time series transformer, where 16% of datasets were used for testing while 84% were used for training. To verify the adequacy of the datasets for training, a "live classification" script was developed to evaluate the Daimon sensor's real-time capability to accurately identify the material.

The following sets of materials were tested:

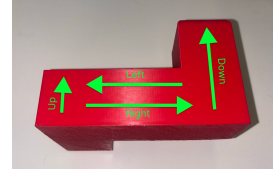
- 3D Plastic Object
- Metal
- Rubber

During the execution of Baxter's movement, the orientation of the object was changed four times. The swiping motion in varying directions is shown in Figure 1. The varying orientations were conducted to analyse the effect of the grain orientation of the material and how that information adds to the material's "fingerprint" and classification. This approach provides a wider range of data that the model is trained on. Without this information, it will not be well-informed, and the accuracy of detection in cases where material orientation is altered will lower predictions' accuracy.

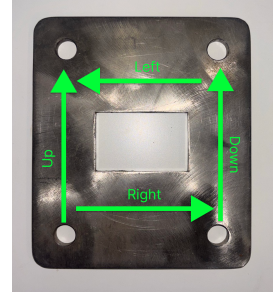
A. Daimon Sensor

The Daimon sensor provides four maps:

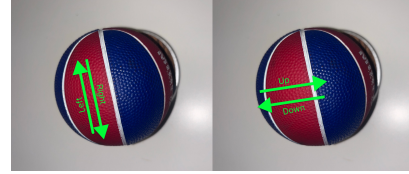
- Raw Image



(a) Plastic object used for testing



(b) Metal object used for testing



(c) Rubber object used for testing

Fig. 1: Materials used for testing with the respective swiping motion indicated

- `getDeformation` - represents two-dimensional deformation of each pixel in mm.
- `getDepth` - represents normal deformation.
- `getShear` - represents tangential deformation

Only two of the four modalities from the Daimon sensor were utilised. The raw image was recommended against for practical use or training in neural networks but is suitable for wear-and-tear analysis, according to the Daimon manual. The primary modalities used by the Daimon sensor were `getDepth` and `getShear`. The `getDepth` modality facilitated elastic deformation perpendicular to the contact surface, whereas the `getShear` modality enabled tangential deformation, which occurs parallel to the contact surface. These two modalities were primarily used to facilitate the implementation of stick-slip logic to distinguish materials.

The derivatives of `getDepth` and `getShear` were also computed. The derivative of `getShear` yielded the rate of tangential variation in the sensor. This value reflects the sensor's slippage on the material and offers information about surface roughness. The derivative of `getDepth` yielded the rate of normal deformation. This value indicates the point at which the material exhibits reduced influence on the sensor, resulting in the deformation of the object rather than the elastomer pad.

This information offers insight on the hardness of materials.

These features were developed for a multi-feature vision-based tactile system analysis.

B. Data Collection

The objective of repeatability is to guarantee that the procedures performed for each material occurred under near-identical conditions. Repeatability led to the incorporation of Baxter to execute the swiping motion. The consistent parameters were the same velocity and applied pressure throughout the swiping motion. This approach enables the comparison of data under comparable settings and, therefore, emphasises the differentiation of materials based on surface roughness and hardness. The absence of repeatability limits the assessment of results necessary to determine which variables affected specific findings and how alterations in those variables could lead to different outcomes.

To maintain a consistent velocity for each object, a time of two seconds was allocated for each swipe. To maintain a consistent beginning pressure, the sensor's initial configuration relative to the object was standardised across all test sets. Baxter's joint mechanism, which employs springs, produces fluctuating pressure as it moves across the material. This information is beneficial if Baxter's applied force is known. Consequently, to enhance understanding of the connection among applied pressure, friction, and sliding velocity, it is imperative to use a force sensor in future experiments.

To execute the swiping command, scripts from Rethink Robotics' GitHub relating to their Baxter repositories were adapted into compatible ROS2 scripts [10], [9], [8], [7], [11], and a web socket algorithm was employed to connect with Baxter.

The packages utilized were the following:

- 1) baxter_interface
- 2) baxter_dataflow
- 3) baxter_simulator
- 4) baxter_common_ros2
- 5) baxter_rosbridge_adapter

The provided scripts assisted in the development of the following scripts:

- ik_baxter: inverse kinematics
- position_kinematics: forward kinematics

This enabled the experimenters to alter the initial position of Baxter's arm with respect to the object. It also assisted in the experimenters' attempt to keep a constant initial applied pressure for each material tested. Upon concluding the experiment, the gathered data were input into the time-series transformer to evaluate the adequacy of the dataset in identifying key material distinctions.

VI. RESULTS AND DISCUSSION

The data acquired from the Daimon sensor included depth and shear, whereas the processed data consisted of shear velocity and depth velocity. The data obtained reflect the

sensor's response to various materials. The effect is influenced by the direction of the applied motion, the magnitude of the pressure exerted, and the surface roughness and hardness of the material.

For each tested material, sixty repetitions were performed across four orientations. The Daimon sensor employs pixels per millimetre as its unit of measurement, which was utilised for analysis.

To sustain continuous motion and applied pressure from Baxter, the object was rotated while maintaining Baxter's fixed location. This method produced varying contact surfaces, particularly for plastic, due to its irregular shape.

A. Analysis of Plastic:

The irregular shape and contact surface area of the plastic object subjected it to different conditions. As illustrated in Figure 1a and Figure 3 the "Plastic up" graph's contact area was less than in the "Plastic down" graph; the difference is indicated by the higher number of pixels per mm.

The longer contact period allowed Baxter to apply varying forces from the start to the end of the motion, which facilitated a more comprehensive analysis of the effects of plastic on the sensor.

The "plastic left" graph and "plastic right" graphs show longer contact with the surface of the object. This data correlates with the illustration of the direction of motion shown in Figure 1.

Shear velocity is the slippage of the sensors. The increased depth, caused by the sensor being pushed into the object, also affected the tangential deformation. It is observed that sliding along the plastic material from the "right" direction resulted in a greater stick-slip effect.

The depth velocity for all directions remained fairly the same; this phenomenon is anticipated because it is a rigid object.

The variability in contact area and pressure, along with the significant influence of the texture of plastic, provided a substantial benefit for training the model by exposing it to diverse circumstances.

B. Analysis of Metal:

The "metal up" and "metal down" graphs illustrated in Figure 4 displayed similar results across all four sections. The data highlights the homogenous grains of metal.

The "metal left" and "metal right" graphs differ among the four sections. The main difference is that a higher pressure may have been applied during the "metal left" test, as indicated by the greater effects and contact points observed in the depth, shear, and shear velocity graphs.

The main difference between the up/down and left/right swiping directions is that when the sensor slid over the metal, it slid over the holes (depicted in Figure 4), which may have produced changes in depth and depth velocity graphs. The last peak on all graphs occurs when the sensor detaches from the material after high pressure has been applied. However, the overall profile depicted remains fairly the same, just at

different scales. The spaced shear velocity indicates that the slippage of the sensor on the metal occurs less frequently, which aligns with the surface roughness of the metal object.

Both depth velocities for the plastic and metal graphs remain fairly consistent throughout the motion and assist in the change of shape of the object; however, this graph changes more drastically when subjected to softer objects.

C. Analysis of Rubber:

Each section displayed in Figure 5 yields a different graph. This highlights the differences in texture along the various axes.

The key difference that contributes to material hardness is the depth of the velocity graphs. This difference differentiates between metal and plastic because the effect of the rubber ball on the sensor changes throughout the swiping motion. Therefore, it reaches a stage where the ball has less of an effect on the sensor than the sensor has on the ball.

The shear-graph contact pixels also vary when compared to plastic and metal. The shear graphs for plastic and metal increase due to increased contact pressure, and the phenomenon is seen via the "spikes" of contact pixels. While in the shear graphs of rubber, the profile of the graph is more curved, which is attributed to the ball's deformation and texture.

The slippage (tangential velocity graphs) is inconsistent and reveals a different understanding than rigid objects. This indicates that the elastomer of the sensor did not "stick-slip" as frequently as it did with metal and plastic.

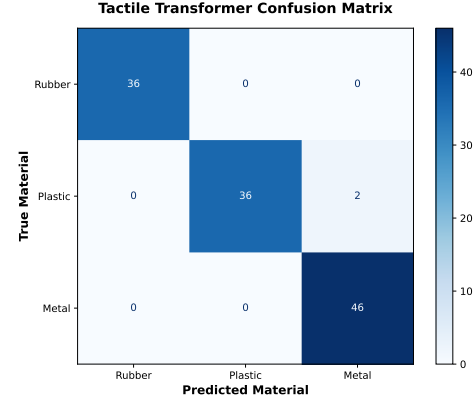
To effectively examine the generated graphs, one should consider the surface finishes of the material, the applied load and velocity of the sensor, and the characteristics of the elastomer pad. The reason for this is that the coefficient of friction is a dynamic system response [5], [6] that is altered by applied normal pressure and sliding velocity. Therefore, the Daimon sensor considers how the elastomer is affected by its perception of the material's surface topology, which is altered by variations in the applied load and velocity. Consequently, increased variability in the data fed into neural networks can enhance comprehension of the material's surface topology for better classification.

D. Neural Network:

The data obtained from the studies were input into a time-series transformer. Figure 2 indicates that the classification report and confusion matrix yielded excellent results, with high weighted and macro averages for precision, recall, and F1-score. This indicates that there are sufficient variances – depth, shear, tangential velocity, depth velocity at applied pressure, and velocity – between the datasets to differentiate the materials effectively.

The confusion matrix assessed the accuracy of the tested data in relation to the actual material. Rubber dominated generally, while plastic and metal had small inaccuracies. A total of 600 files were used for training, whereas 120 files were employed for testing. The dataset is naturally limited for training, as it bounds metal, plastic, and rubber classification

to these specific objects. This limitation includes the lack of subtypes of these materials (i.e., the different hardness between ABS and PET), nor does it consider the diverse surface finishes due to manufacturing methods. However, the outcomes obtained are relatively favourable, indicating a clear relationship between the materials using the Daimon sensor.



(a) Confusion Matrix for the evaluation of the test dataset

	Precision	Recall	F1-Score	Support
Rubber	1.00	1.00	1.00	36
Plastic	1.00	0.95	0.97	38
Metal	0.96	1.00	0.98	46
Accuracy			0.98	120
Macro Avg	0.99	0.98	0.98	120
Weighted Avg	0.98	0.98	0.98	120

(b) Classification Report of the trained data

Fig. 2: Overall evaluation for the trained and tested data

E. Live Classification:

In the real-time manual human testing, the materials evaluated were those utilised in the data-gathering experiments. The live classification script utilised the same architectural model of the neural network and collected data through the same methods employed during experimental tests. The trained data from the neural network were fed into the classifier so as to compare the live testing against what the model was trained on.

As previously analysed, the results from the neural network obtained favourable analytical results, which suggested that a discrepancy existed among the various materials. However, when subjected to real-time testing, the system inaccurately classified the tested material. Approximately 70% of misclassifications identified the live-tested item as "plastic". The misclassification may be attributed to the fact that plastic underwent a broader range of testing conditions. This variation included inconsistent sensor-applied pressure over a two-second duration due to the irregular shape of the item. This variation is better indicated when referencing Figure 3. The difference in contact area between the "right" direction and

the "up" direction indicates that the contact time was shorter for the "up" direction; therefore, it did not achieve the same level of applied pressure, contact time, or frequency of stick-slip transitions as the "right" direction. While rubber and metal were consistently exposed to identical conditions, any deviation from these parameters led to misclassifications. This results from the architecture of the neural network and its method of classifying the experimental profiles. It was unable to distinguish between the applied pressures throughout the swiping process, except when tested in a different set of trials, such as the experiments involving plastic.

VII. CONCLUSION:

The inclusion of applied pressure along the motion of the sensor at a constant velocity proved to be useful in differentiating the chosen materials, as evaluated by Figure 2a.

During the live classification tests, it became evident that discrepancies in testing pressure and velocity compared to the experimental conditions affected the classification system's ability to appropriately identify the material. The justification for this discrepancy is that the trained model does not consider the explicit behaviour of the elastomer pad under different velocities and applied pressures, particularly for metal and rubber. Improving the methodology for testing datasets will yield a more accurate "fingerprint" of a material.

An additional constraint is the evaluation on non-experimental objects; the classifier could only select from the readily trained materials: plastic, metal, or rubber. Consequently, the insufficient variety of materials and their classifications resulted in misclassifications.

To train a more robust multi-modal classification model, more experiments need to be conducted under varying normal load and sliding velocities, as well as across various material categories and subcategories. Subcategories encompass various manufacturing procedures for the materials (e.g., molded plastic versus 3D-printed plastic). With this approach, the obtained data will provide a better "fingerprint" of each material, which will lead to a more accurate and diverse classifier. To comprehend the limitations and conduct tests effectively, it is also necessary to consider how the coefficient of friction changes with respect to contact materials, applied pressure, and sliding velocity, and to potentially establish a baseline comparison utilizing finite element analysis models. The reason for this approach is to fully exploit the capabilities of the Daimon sensor and to better anticipate the implications of the produced data.

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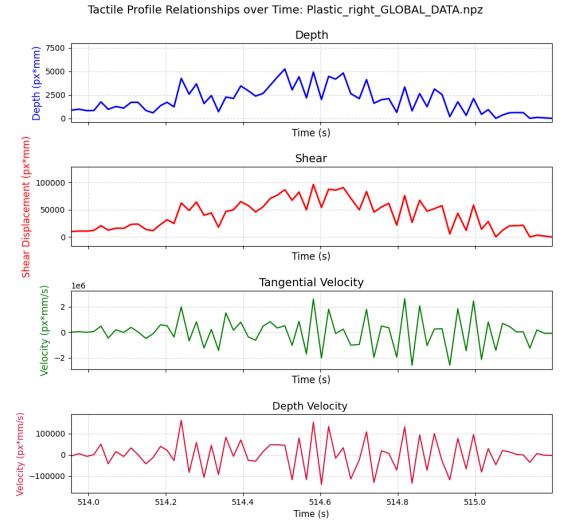
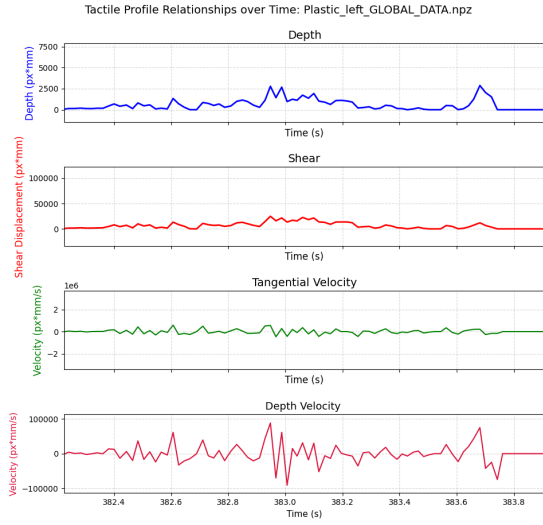
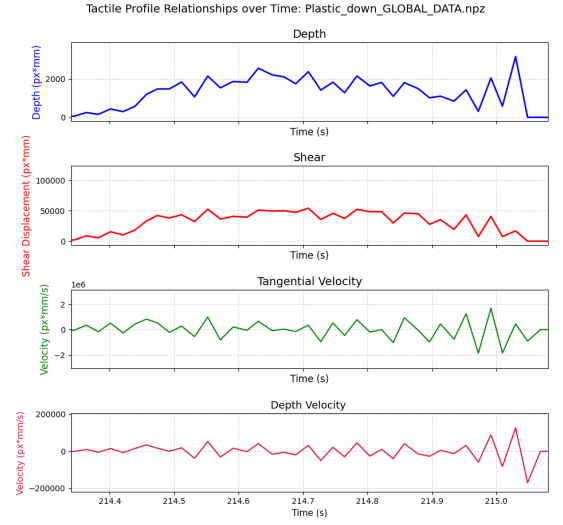
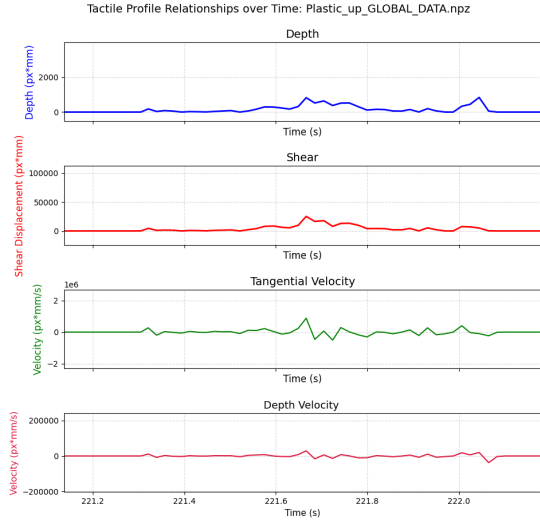


Fig. 3: A section taken from the sensor sliding over plastic in the four directions.

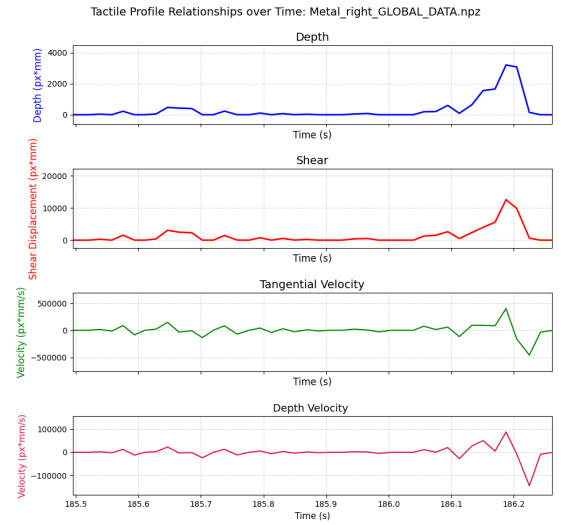
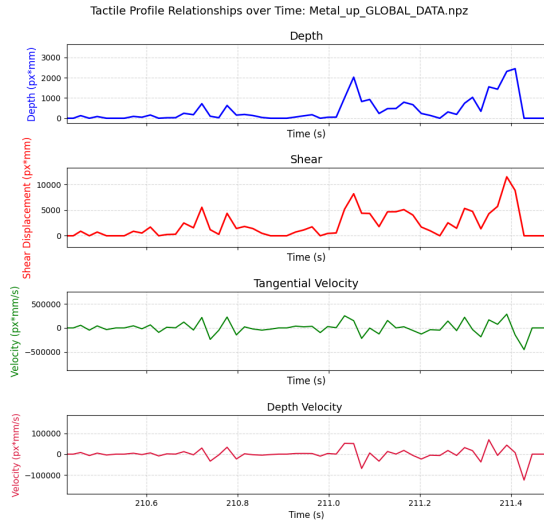


Fig. 4: A section taken from the sensor sliding over metal in the four directions.

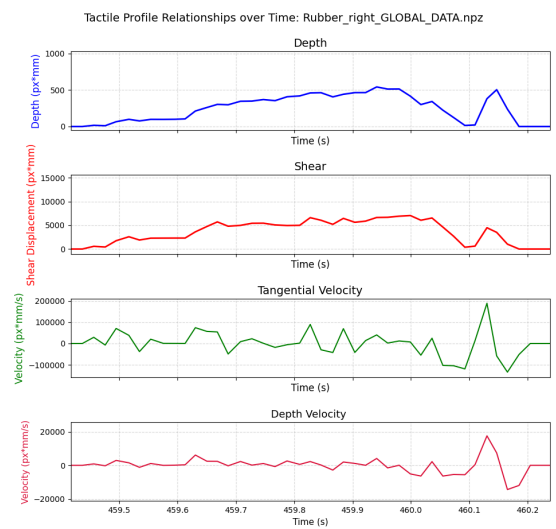
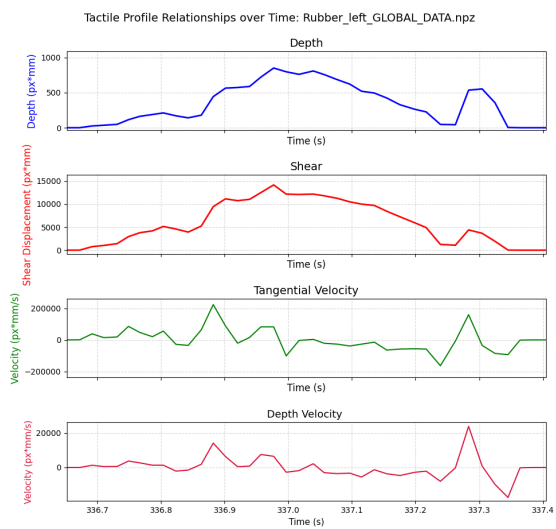
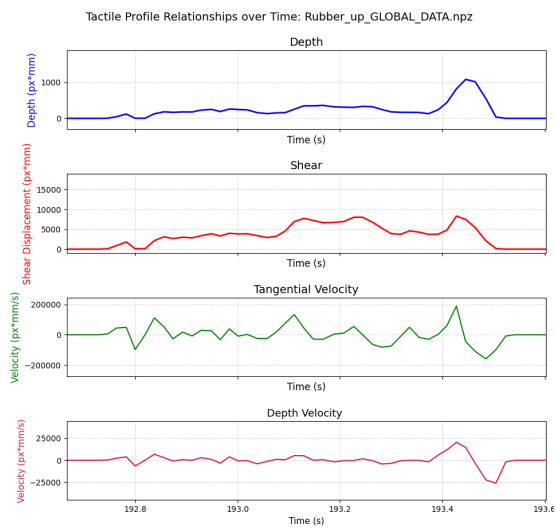


Fig. 5: A section taken from the sensor sliding over rubber in the four directions.